

Project Design Phase

Solution Architecture

Field	Details
Date	28 September 2026
Team ID	SWTID-2026-4282
Project Name	Streamlining IT Procurement: Automating Standard Laptop with flow Designer
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Project Architecture Components & Specifications

1. Core Architecture Components

- **Presentation Layer (User Interface):** ServiceNow Service Portal / Employee Center where users submit the 'Standard Laptop Request' catalog item.
- **Orchestration & Logic Layer (Central Brain):** ServiceNow Flow Designer which executes the workflow, data transitions, conditional routing, and error handling.

- **Integration Layer:** ServiceNow Integration Hub utilizing REST APIs and predefined Spokes to communicate with external applications.
- **Data & Asset Layer:** ServiceNow ITAM (IT Asset Management) Configuration Management Database (CMDB) to track hardware models, stock quantities, and serial numbers.

2. External Integration Endpoints

- **HR System Integration:** Workday / SuccessFactors API to fetch new hire profiles and auto-trigger laptop requests.
- **Communication Channel Integration:** Microsoft Teams / Outlook Spoke to send interactive adaptive cards for manager approvals.
- **Vendor Procurement Integration:** Hardware Vendor B2B API (e.g., Dell, HP, or Lenovo) to programmatically dispatch Purchase Orders (POs) when internal stock is unavailable.

Data Flow Description (For Architecture Diagram)

1. **Trigger Event:** Employee requests a laptop via the Service Portal OR an automated HR profile sync creates a request trigger.
2. **Evaluation Phase:** Flow Designer queries the local ServiceNow Inventory Table to check if the standard laptop model is currently in stock.
3. **Approval Node:** Flow Designer initiates a parallel workflow step sending a dynamic approval notification to the designated Manager via Microsoft Teams Spoke.
4. **Conditional Branching:**
 - *Branch A (In Stock):* If approved and stock is present, Flow Designer allocates the asset tag, updates CMDB status to 'Reserved', and triggers a deployment task for the IT technician.
 - *Branch B (Out of Stock):* If approved but stock is empty, Integration Hub triggers an outbound REST API Call to the Vendor Portal to auto-generate a hardware Purchase Order (PO).
5. **Closure & Tracking:** The system updates the end-user with a tracking number via email notification and updates the ticket status to 'Fulfilled'.